

Mobile Application Development with Flutter

Lecture 37. GridView Widget in Flutter

What is GridView Widget?

- GridView is a scrollable widget in Flutter that displays items in a two-dimensional layout (rows and columns)
- It's commonly used for layouts like photo galleries or product catalogs

Key Characteristics

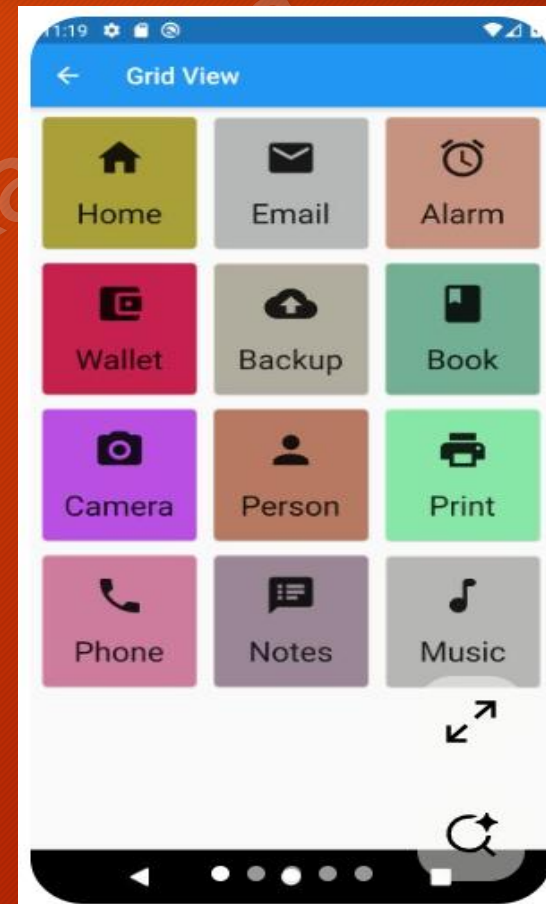
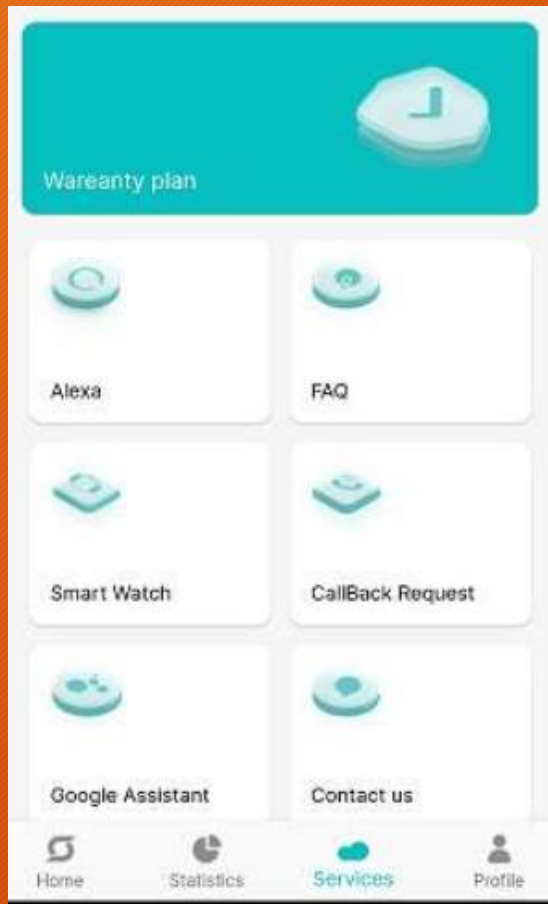
- Displays items in **grid format**
- Supports **vertical scrolling** (default)
- Can be **static or dynamic**
- Efficient when used with `.builder`

Real-World Use Cases

-  E-commerce apps (products)
-  Image galleries (Instagram, Google Photos)
-  Movie apps (Netflix thumbnails)
-  App launchers (icons grid)

Prepared by Ehtisham Rasheed

Some Screenshots





When to Use GridView

Use GridView when:

Situation	Widget
Single column list	ListView
Multiple columns needed	GridView
Same type of repeated items	GridView.builder
Dynamic data (API)	GridView.builder

GridView.count() Constructor

- This is used for small, static grids where the number of items is known beforehand
- It requires a `crossAxisCount` property to specify a fixed number of columns

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GridView.extent() Constructor

- Similar to `GridView.count()`, but instead of a fixed count, it allows you to specify a `maxCrossAxisExtent` (maximum width) for each tile
- The number of columns is automatically calculated based on the available space
- It is useful for responsive designs

GridView.builder() Constructor

- This is the most efficient constructor for large or dynamically changing datasets
- It only builds the widgets that are currently visible on the screen (lazy loading)
- It requires an itemBuilder function and a gridDelegate property

Prepared by Eniasham Rashid

GridView.custom() Constructor

- This provides full control over the grid layout and child management using custom SliverGridDelegate and SliverChildDelegate

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Important Properties

- `crossAxisCount`
 - Number of columns
- `mainAxisSpacing`
 - Vertical space between items
- `crossAxisSpacing`
 - Horizontal space
- `gridDelegate`
 - Controls layout behavior

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